Devixo Solutions

AI/ML Internship Program

Task 03 – End-to-End Machine Learning Project

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1. Introduction

This project develops a machine-learning workflow to estimate the price of a used vehicle from available vehicle details. The work includes dataset inspection and cleaning, feature engineering, preprocessing, training and comparing regression models, hyperparameter tuning, and building a Streamlit application for interactive predictions.

2. Objective

- Prepare and clean a used-vehicle dataset for regression modelling.
- Create useful features from vehicle year and engine specifications.
- Train and compare multiple regression models using consistent evaluation metrics.
- Tune a Random Forest model and compare its results with the baseline models.
- Select a model based on measured performance and use it in a Streamlit application.
- Document the model's current limitations and identify practical future improvements.

3. Dataset Name and Source

Dataset: Used Car Price Prediction Dataset

Source: Kaggle

URL: https://www.kaggle.com/datasets/taeefnajib/used-car-price-prediction-dataset

Note: The currency associated with the dataset's price values has not been independently verified. The currency symbol used in the application should be checked against the original dataset source.

4. Dataset Description

The dataset contains vehicle attributes and a numeric target, price, making this a supervised regression problem. The working cleaned dataset used for modelling contains 4,009 rows and 16 columns. The original fields describe vehicle make and model, year, mileage, fuel, engine, transmission, colours, accident history, title status, and price.

5. Dataset Size

Stage	Rows	Columns / Notes
Original dataset fields	Not restated here	12 original columns
Working cleaned dataset	4,009	16 columns after feature engineering
Training set	3,207	80% of cleaned data
Test set	802	20% of cleaned data

6. Features / Columns Description

Feature / Column	Description
brand	Vehicle manufacturer or brand.
model	Vehicle model name.
model_year	Model year listed in the dataset.
milage	Recorded vehicle mileage; converted to numeric form.
fuel_type	Fuel type category.
engine	Text description of the vehicle's engine.
transmission	Transmission category or description.
ext_col	Exterior colour.

int_col	Interior colour.
accident	Recorded accident-history category.
clean_title	Clean-title status category.
price	Numeric target variable to be predicted.
car_age	Engineered vehicle age calculated from model year.
engine_liters	Engine displacement extracted or derived from engine text.
horsepower	Horsepower extracted from engine text when available.
cylinders	Cylinder count extracted from engine text when available.

7. Methodology

1. Load the dataset and inspect its shape, columns, data types, missing values, duplicates, and summary statistics.
2. Convert string-formatted price and mileage values to numeric values and ensure model year is numeric.
3. Create car_age, engine_liters, horsepower, and cylinders through feature engineering.
4. Split the cleaned data into 80% training data and 20% test data using random_state = 42.
5. For numeric features, use median imputation and StandardScaler. For categorical features, use constant imputation with the value Missing and one-hot encoding with unknown categories ignored.
6. Train and compare Linear Regression, Ridge Regression, Decision Tree, Random Forest, and Gradient Boosting.
7. Tune a Random Forest model using RandomizedSearchCV with five-fold cross-validation and a log-transformed target.
8. Evaluate models with MAE, RMSE, and R^2, then save the selected baseline Random Forest model for the Streamlit application.
9. Run the Streamlit application locally to accept vehicle inputs and display an estimated price.

8. Data Loading

The dataset was loaded into a pandas DataFrame for analysis and modelling. Pandas and NumPy were used for data handling, while scikit-learn was used for preprocessing, regression models, model evaluation, and hyperparameter search. The application uses Streamlit and loads the saved model from used_vehicle_price_model.joblib.

9. Data Cleaning and Feature Engineering

- The initial inspection found 170 missing values in fuel_type, 113 in accident, and 596 in clean_title.
- No duplicate rows were found in the initial duplicate check.
- The price and milage fields were converted from string-formatted values to numeric values.
- The car_age feature was calculated as (2026 − model_year), with negative values clipped to zero.
- Engine text was used to derive engine_liters, horsepower, and cylinders where information could be extracted.
- Numeric features were prepared with median imputation and scaling; categorical features were imputed and one-hot encoded.

10. Exploratory Data Analysis (EDA)

EDA was used to inspect data structure, missing values, duplicates, feature types, and price distribution. The price values are strongly right-skewed: the median was 31,000, the mean was 44,553, the 99th percentile was 272,713.28, and the maximum was 2,954,083. This indicates that a small number of high-priced vehicles can strongly influence error metrics.

11. Statistical Analysis and Evaluation Metrics

The models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2. MAE represents the average absolute difference between predictions and actual prices. RMSE gives greater weight to larger errors. R^2 measures model performance relative to a simple mean-based baseline.

The price distribution statistics for the working dataset were: median = 31,000; mean = 44,553; 99th percentile = 272,713.28; and maximum = 2,954,083. Currency is not specified here because it has not been independently verified.

12. Model Comparison Results

Model	MAE	RMSE	R^2
Random Forest	17,192.81	133,047.05	0.1340
Gradient Boosting	20,170.42	133,351.76	0.1300
Decision Tree	20,455.82	135,396.01	0.1031
Ridge Regression	22,114.16	135,600.36	0.1004
Linear Regression	22,251.00	138,818.00	0.0572

The baseline Random Forest had the lowest MAE and RMSE and the highest R^2 among the five initial configurations, so it was retained as the selected model for the app.

13. Hyperparameter Tuning Results

A separate RandomizedSearchCV run evaluated a Random Forest with a log-transformed target. The best cross-validation RMSE reported was 29,494.39. The best parameters were n_estimators = 100, min_samples_split = 5, min_samples_leaf = 1, max_features = 1.0, and max_depth = 20.

Evaluation	MAE	RMSE	R^2
Tuned log-target Random Forest — test set	17,283.71	135,127.22	0.1067

The tuned model did not outperform the baseline Random Forest on the reported test metrics. The baseline model was therefore retained for the Streamlit application. The cross-validation RMSE and test RMSE are produced by different evaluation procedures and should not be treated as directly equivalent.

14. Major Findings and Feature Importance

- Random Forest performed best in the initial model comparison, although its test R^2 was 0.1340.
- The tuned log-target Random Forest did not improve on the baseline model's reported test metrics.
- Mileage had the highest reported feature-importance score (0.3315).
- Engine litres (0.1242), Lamborghini brand indicator (0.1073), Rolls-Royce brand indicator (0.1020), and horsepower (0.0800) were also among the leading features.
- Feature-importance scores describe how the model used features; they do not establish causation or represent percentages of price explained.

15. Ten Insights

i. The working cleaned dataset contained 4,009 rows and 16 columns after feature engineering.
ii. The original dataset had 12 fields before engineered features were added.
iii. Missing values were present in fuel_type, accident, and clean_title.
iv. No duplicate rows were found in the initial duplicate check.

v. Price and mileage needed conversion from string-formatted values to numeric values.
vi. The data was split into 3,207 training rows and 802 test rows.
vii. The median price was 31,000, while the mean was 44,553, showing a right-skewed distribution.
viii. Random Forest had the lowest MAE among the five initial models (17,192.81).
ix. Random Forest also had the highest R^2 among the initial comparison models (0.1340), but the score still indicates limited predictive fit.
x. The tuned log-target Random Forest did not outperform the baseline Random Forest on the reported test metrics.

16. Limitations

- The baseline Random Forest test R^2 of 0.1340 indicates that the model explains only a limited share of price variation.
- A small number of very expensive vehicles creates a highly skewed target distribution and can increase RMSE.
- The dataset may lack important pricing information such as trim, vehicle condition, location, service history, seller type, and market timing.
- Engine specifications extracted from text can be incomplete or inconsistent.
- The model may not generalize to different markets or time periods.
- The test split was used during model comparison and selection, so the reported results are a practical benchmark rather than a completely untouched final evaluation.
- The price currency has not been independently verified and should be checked before labelling predictions with a specific currency.

17. Future Improvements

- Verify the dataset's currency, units, source, and collection date.
- Investigate extreme prices and test robust approaches to outliers without automatically removing legitimate luxury vehicles.
- Use cross-validation during model selection and reserve a final holdout set for one-time evaluation.
- Test additional models such as XGBoost, LightGBM, or CatBoost if available.
- Improve engine-text parsing and evaluate the effect of missing engineered features.
- Add more relevant variables, including trim, condition, location, service history, and listing date.
- Add prediction intervals or uncertainty estimates to show that outputs are estimates.
- Improve the Streamlit interface with input validation, clear units and currency labels, and example values.
- Evaluate the model on newer listings and retrain it when market conditions change.

18. Conclusion

The project completed an introductory used-vehicle price prediction workflow, including data cleaning, feature engineering, preprocessing, model comparison, tuning, and a local Streamlit application. The baseline Random Forest performed best among the initial models, but its R^2 score indicates that the model requires further improvement before predictions should be relied on for real purchasing decisions.

19. GitHub Repository Link

Repository URL: https://github.com/shafaque-shahid/Devixo_Solutions_Task03